

Yongjun Rong

rongyj@hotmail.com | 856-408-4027 | [linkedin.com/in/rongyongjun](https://www.linkedin.com/in/rongyongjun) | github.com/rongyj
Cherry Hill, NJ, USA, 08003 | US Citizen

Summary

A Senior Software Engineer with deep expertise in building Core GenAI Platforms and LLM infrastructure. I specialize in designing and implementing scalable systems for model inference, prompt orchestration, and RAG workflows using Python, FastAPI, and modern AI toolchains. With experience across multiple organizations, I have built platforms that democratize access to generative AI through clean APIs, reusable components, and intuitive tooling. My work focuses on creating reliable, observable infrastructure that enables cross-functional engineering teams to adopt and scale GenAI capabilities while ensuring performance, security, and compliance.

Tags: GenAI Platform, LLM Infrastructure, Senior Software Engineer

Skills

GenAI Platform Development: Extensive experience building core GenAI platforms with LLM inference systems, prompt orchestration, and RAG workflows using Python, FastAPI, and modern AI toolchains (LangChain, OpenAI API).

Scalable AI Infrastructure: Proven ability to design and implement reliable, observable infrastructure for model inference, data workflows, and AI orchestration using Kubernetes, Docker, and cloud platforms (AWS).

Cross-Functional Platform Leadership: Experience driving adoption of AI tooling across engineering teams through reusable components, clean APIs, and intuitive developer experiences. Strong stakeholder management and cross-functional collaboration skills.

Full-Stack Engineering: Proficiency in backend development with Python/FastAPI and microservices architecture, with experience in infrastructure tools like Terraform, Kubernetes, and observability systems (Prometheus, Grafana, CloudWatch).

Production AI Operations: Hands-on experience operationalizing LLMs in production, including latency management, prompt engineering, vector databases, and ensuring system reliability through metrics-driven approaches.

ML Platform Infrastructure: Familiarity with data processing pipelines, model inference optimization, and building platforms that support machine learning workflows at scale.

Technologies

AI Platform Tools: Python/FastAPI/Pydantic, OpenAI API, LangChain, RAG, LLM Inference, Vector Stores, Prompt Engineering

Infrastructure Tools: Kubernetes/EKS, Terraform/Terragrunt, AWS (Lambda, SQS, SNS, EventBridge), Docker, Observability (CloudWatch, Prometheus)

Programming Languages: Python, Java/Kotlin, TypeScript, Terraform

Work Experience

Director of Cloud Architecture

Feb. 2025 – Current

Biophy, Cherry Hill, NJ (WFH), USA

Architected and implemented a comprehensive GenAI platform with RAG capabilities, designing the core infrastructure for model inference, prompt orchestration, and scalable data workflows. Built a unified interface for secure LLM interactions using modern AI toolchains.

Technologies: Python/FastAPI/Pydantic, OpenAI API/LangChain, RAG, LLMs, AWS/Lambda/SQS/SNS, Infrastructure as Code, Kubernetes, Observability (CloudWatch)

Project: AI-Powered Data Ingestion and Scraping System for RAG

- Implemented FastAPI services to expose LLM capabilities as clean, reusable APIs for cross-functional teams.
- Architected modular data ingestion pipeline supporting multiple data sources and formats for RAG systems.
- Designed prompt orchestration layers following clean architecture principles for LLM interactions.
- Built robust error handling and observability mechanisms for AI workflow failures.

Senior Cloud Engineer II (Associate)

Dec. 2021 – Feb. 2025

McKinsey & Company, Cherry Hill, NJ (WFH), USA

Built and scaled GenAI systems with RAG and multi-agent frameworks, developing core platform infrastructure for LLM inference, vector databases, and prompt engineering. Led cross-functional adoption of AI tooling across engineering teams.

Technologies: Python/FastAPI/Pydantic, OpenAI API/LangChain, RAG/Multi-agents, VectorStore/ChromaDB, Kubernetes/EKS, Terraform, Observability, Microservices

Projects:

- **GenAI System with RAG for Accelerating Data Pipeline Development**
 - Implemented FastAPI APIs to expose AI capabilities as reusable services for engineering teams.
 - Designed and implemented chunking systems for LLM context window management.
 - Built parallel inference calls to OpenAI API using asyncio for performance optimization.
 - Developed prompt tuning systems for code generation and unit test creation.
 - Created developer tooling and documentation for local development and debugging workflows.
- **Multi-Agent GenAI System for Mainframe Migration**
 - Developed multi-agent GenAI systems for automated development/testing/deployment processes.
 - Implemented RAG systems with vector stores for context-aware agentic systems.
 - Designed scalable inference infrastructure for AI workflow orchestration.
 - Built observability and monitoring for AI system performance and reliability.

Staff Engineer

Dec. 2020 – Dec. 2021

Yotyscale, Cherry Hill, NJ (WFH), USA

Led architecture and development of a cloud analytics platform, building scalable data pipelines, microservices, and infrastructure with strong observability and metrics-driven approaches.

Technologies: Python/Java/Kotlin, Kubernetes/EKS, Terraform/Terragrunt, Microservices, Observability, CI/CD, Data Pipelines

Project: Cloud Cost Analytics Platform Architecture

- Designed and developed analytics microservices with ClickHouse and PostgreSQL.
- Implemented CI/CD pipelines for microservices deployment to EKS.
- Built comprehensive observability with metrics and monitoring.
- Developed Terraform modules for infrastructure automation.
- Created dynamic metadata-driven configuration systems.

Education

Master of Computer Science

Texas Tech University, Lubbock, TX, USA

GPA: 4.0/4.0

Sept. 2002 – Jun. 2006

Languages

English: Fluent

Chinese: Native

Hobbies

Playing with cutting-edge technologies: genAI/LangChain/Multi-agents, Docker, Kubernetes, AI Infrastructure

iOS/Android Apps Development: Developing applications with AI integration and modern toolchains